UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

STAT 576: Empirical Process Theory and Weak Convergence — Fall 2025
Lecturer: Eren C. Kızıldağ — Sep 9, 2025
Scriber(s): Tianyang Jiang

Lecture 5: Stein's Method for Dependent R.V.'s

Definition 5.1 A graph is a pair $G = (V, E)$ consisting of a set of vertices V and a set of edges $E \subseteq \{(v, v') : v, v' \in V,\ v \neq v'\}$. We consider undirected graphs.

Definition 5.2 For $v \in V$, let the (open) neighborhood be

$$N_v := \{\, v' \in V : (v, v') \in E \,\}, \qquad d_v := |N_v| \ \text{ (the degree of } v\text{)}.$$

The maximum degree is

$$\Delta := \max_{v \in V} d_v.$$

Setup: Let $\{X_v : v \in V\}$ be a collection of random variables indexed by V.

Definition 5.3 A graph $G = (V, E)$ is a *dependency graph* for the family $\{X_v : v \in V\}$ if for any disjoint $S, T \subseteq V$ such that there is *no* edge with one endpoint in S and the other in T (i.e. $(s, t) \notin E$ for all $s \in S, t \in T$), the sub-families $\{X_s : s \in S\}$ and $\{X_t : t \in T\}$ are independent.

Example 5.4 If $E = \varnothing$ (the empty graph), then G is a dependency graph for any collection of *independent* random variables $\{X_v : v \in V\}$.

Example 5.5 Let $Y_1, \ldots, Y_n$ be independent. Define $X_i := Y_i Y_{i+1}$ for $i = 1, \ldots, n-1$. Then a dependency graph for $\{X_1, \ldots, X_{n-1}\}$ is the path graph $1 - 2 - \cdots - (n-1)$: X_i is dependent only on its neighbors X_{i-1} and X_{i+1} (through Y_i and Y_{i+1}), while X_i and X_j are independent whenever $|i - j| > 1$.

Intuition: Let $D := 1 + \Delta$ (recall $\Delta = \max_{v \in V} d_v$). When D (equivalently, the maximum degree Δ) is small, then we get a CLT.

Lemma 5.6 Let D be defined as above for dependency graph of $\{X_i\}$, then we have

$$\text{VAR}\left(\sum_{i=1}^{n} X_i\right) \leq D \sum_{i=1}^{n} \text{VAR}(X_i)$$

Proof. Let $\mathbb{E}X_i = 0$ without loss of generality.
Let $N_i := \{j : (i,j) \in E\}, \qquad \overline{N}_i := N_i \cup \{i\}.$

$$\mathbb{E}\sum_i X_i = 0 \;\Rightarrow\; \text{VAR}\Big(\sum_i X_i\Big) = \mathbb{E}\left[\Big(\sum_i X_i\Big)^2\right],\ \mathbb{E}[X_i X_j] = 0, \text{ if } j \notin \overline{N}_i.$$

Expanding,

$$\begin{aligned}\mathbb{E}\left[\Big(\sum_i X_i\Big)^2\right] &= \mathbb{E}\left[\sum_i \sum_j X_i X_j\right] = \sum_i \sum_{j \in \overline{N}_i} \mathbb{E}[X_i X_j] \\ &\leq \sum_i \sum_{j \in \overline{N}_i} \mathbb{E}\left[\frac{X_i^2 + X_j^2}{2}\right] \leq D \sum_i \text{VAR}(X_i),\end{aligned}$$

where we used the AM-GM inequality $x^2 + y^2 \geq 2|xy|$ valid for all $x, y \in \mathbb{R}$.

Theorem 5.7 Let $X_1, \ldots, X_n$ be zero-mean RVs with

$$\sigma^2 = \text{VAR}\Big(\sum_{i=1}^{n} X_i\Big).$$

Define $W = \frac{1}{\sigma}\sum_{i=1}^{n} X_i$. Then

$$d_{\text{W}}(W, Z) \;\leq\; \frac{4}{\sigma^2\sqrt{\pi}} \sqrt{D^3 \sum_{i=1}^{n} \mathbb{E}|X_i|^4} \;+\; \frac{D^2}{\sigma^3} \sum_{i=1}^{n} \mathbb{E}|X_i|^3,$$

where D is as above.

Proof.

We begin by recalling the key ingredient of Stein's method.

Theorem 5.8 We have

$$\begin{aligned} d_{\mathrm{W}}(W,Z) &= \sup_{g:g \text{ is 1-Lipschitz}} \big|\mathbb{E}g(W) - \mathbb{E}g(Z)\big| \\ &= \sup_{f\in\mathcal{D}} \big|\mathbb{E}f'(W) - \mathbb{E}Wf(W)\big|, \end{aligned}$$

where

$$\mathcal{D} := \{f : \mathbb{R}\to\mathbb{R} : \|f\|_\infty \le 1, \|f'\|_\infty \le \sqrt{2/\pi}, \|f''\|_\infty \le 2\}.$$

Let $\overline{N}_i$ be as above ($\overline{N}_i = \{i\} \cup N_i$). Set $W_i = \frac{1}{\sigma}\sum_{j\in\overline{N}_i} X_j$.
Take any $f : \mathbb{R}\to\mathbb{R}$ with

$$\|f\|_\infty \le 1, \qquad \|f'\|_\infty \le \sqrt{\frac{2}{\pi}}, \qquad \|f''\|_\infty \le 2.$$

We proceed to control $\big|\mathbb{E}f'(W) - \mathbb{E}[Wf(W)]\big|$.

$$\begin{aligned} \mathbb{E}[Wf(W)] &= \frac{1}{\sigma}\sum_i \mathbb{E}[X_i f(W)] \\ &= \frac{1}{\sigma}\sum_i \mathbb{E}\big[X_i\{f(W) - f(W_i) - (W-W_i)f'(W)\}\big] \\ &+ \frac{1}{\sigma}\sum_i \mathbb{E}\big[X_i(W-W_i)f'(W)\big]. \end{aligned}$$

$$\big(\mathbb{E}[X_i f(W_i)] = 0 \text{ as } X_i \perp W_i\big)$$

Corollary (I).

$$\text{(I)} := \frac{1}{\sigma}\sum_i \mathbb{E}\big[X_i\{f(W) - f(W_i) - (W-W_i)f'(W)\}\big].$$

By Taylor expansion,

$$f(W_i) = f(W) + (W_i - W)f'(W) + \frac{(W_i-W)^2}{2} f''(\theta_i), \qquad \theta_i \in (\min(W,W_i), \max(W,W_i)).$$

Hence

$$f(W) - f(W_i) - (W-W_i)f'(W) = \frac{(W-W_i)^2}{2} f''(\theta_i),$$

so

$$\text{(I)} = \frac{1}{\sigma} \sum_i \mathbb{E}\left[\frac{1}{2} X_i (W - W_i)^2 f''(\theta_i)\right].$$

$$W - W_i = \frac{1}{\sigma} \sum_{j \in \overline{N}_i} X_j.$$

Therefore,

$$|\text{(I)}| \le \frac{1}{2\sigma^3} \mathbb{E}\left[\sum_i |X_i| \Big(\sum_{j \in \overline{N}_i} X_j \Big)^2\right] \|f''\|_\infty \qquad (\|f''\|_\infty = 2)$$

$$\le \frac{1}{\sigma^3} \mathbb{E}\left[\sum_i |X_i| \Big(\sum_{j \in \overline{N}_i} X_j \Big)^2\right].$$

By AM–GM for $x, y, z \ge 0$, $xyz \le \dfrac{x^3 + y^3 + z^3}{3}$, hence

$$\frac{1}{\sigma^3} \sum_i \sum_{j \in \overline{N}_i} \sum_{k \in \overline{N}_i} \mathbb{E}\big[\, |X_i|\, |X_j|\, |X_k| \,\big] \;\le\; \frac{1}{\sigma^3} \sum_i \sum_{j \in \overline{N}_i} \sum_{k \in \overline{N}_i} \frac{\mathbb{E}|X_i|^3 + \mathbb{E}|X_j|^3 + \mathbb{E}|X_k|^3}{3}.$$

$$\text{So} \qquad \text{(I)} \;\le\; \frac{D^2}{\sigma^3} \sum_i \mathbb{E}|X_i|^3.$$

Bounding (II). Again,

$$W - W_i = \frac{1}{\sigma} \sum_{j \in \overline{N}_i} X_j.$$

So (II) becomes

$$(II) = \frac{1}{\sigma^2} \sum_i \sum_{j \in \overline{N}_i} \mathbb{E}\big[X_i X_j f'(W)\big].$$

$$\big|\mathbb{E} f'(W) - \mathbb{E}[W f(W)]\big| \le |(I)| + \big|\mathbb{E} f'(W) - (II)\big|.$$

$$\mathbb{E} f'(W) - (II) = \mathbb{E}\left[\left(\frac{1}{\sigma^2} \sum_i \sum_{j \in \overline{N}_i} X_i X_j - 1\right) f(W)\right].$$

Let

$$T := \frac{1}{\sigma^2} \sum_i \sum_{j \in \overline{N}_i} X_i X_j,$$

$$T' := \frac{1}{\sigma^2} \sum_i \sum_{j \in \overline{N}_i} X_i X_j - 1.$$

Then

$$\mathbb{E}T = \frac{1}{\sigma^2} \mathbb{E}\left(\sum_i X_i \right)^2 = 1, \qquad \mathbb{E}T' = 1 \Rightarrow \mathbb{E}[T' - 1] = 0.$$

So,

$$\left| \mathbb{E}f'(W) - (II) \right| \leq \mathbb{E}|T'| \, \|f\|_\infty \leq (\mathbb{E}|T'|) \sqrt{\frac{2}{\pi}},$$

where $T' := T - 1$. Since $\mathbb{E}T' = 0$,

$$\mathbb{E}|T'| \leq \sqrt{\mathbb{E}(T')^2} \quad \text{(by Cauchy–Schwarz)},$$

and $\mathbb{E}(T')^2$ is nothing but the variance of T'.

$$\left| (II) - \mathbb{E}f'(W) \right| \leq \sqrt{\frac{2}{\pi}} \, \sqrt{\text{VAR}(T)}.$$

$$T' = \frac{1}{\sigma^2} \sum_i \sum_{j \in \overline{N}_i} X_i X_j - 1, \qquad \text{VAR}(T') = \text{VAR}(T).$$

Hence, the upper bound becomes

$$\sqrt{\frac{2}{\pi} \, \text{VAR}\left(\frac{1}{\sigma^2} \sum_i \sum_{j \in \overline{N}_i} X_i X_j \right)} \; = \; \frac{1}{\sigma^2} \sqrt{\frac{2}{\pi} \, \text{VAR}\left(\sum_i \sum_{j \in \overline{N}_i} X_i X_j \right)}.$$

$$\{ \, X_i X_j : i = 1, \ldots, n, \; j \in \overline{N}_i \, \}.$$

Consider $X_i X_j$ and $X_k X_\ell$.
These r.v.'s define a new dependency graph with vertices $\{(i, j) : \; i = 1, \ldots, n$ and $j \in \overline{N}_i\}$. $X_i X_j$ and $X_k X_\ell$ are dependent if $k \in \overline{N}_i \cup \overline{N}_j$ or $\ell \in \overline{N}_i \cup \overline{N}_j$. So

$$\left| \overline{N}_i \cup \overline{N}_j \right| \leq |\overline{N}_i| + |\overline{N}_j| \leq 2D.$$

And once k is fixed, the other one has D many choices.

$\Rightarrow$ $2D^2+1$ is the (1+max degree) in the dependency graph of $\{X_iX_j : i = 1, ...n, j \in \overline{N}_i\}$.

Therefore, by the earlier lemma,

$$\begin{aligned}
\text{VAR}\Big(\sum_i \sum_{j\in\overline{N}_i} X_iX_j\Big) &\le (2D^2+1)\sum_i \sum_{j\in\overline{N}_i} \text{VAR}(X_iX_j) \\
&\le (2D^2+1)\sum_i \sum_{j\in\overline{N}_i} \mathbb{E}\big[X_i^2X_j^2\big] \\
&\le (2D^2+1)\sum_i \sum_{j\in\overline{N}_i} \frac{\mathbb{E}X_i^4 + \mathbb{E}X_j^4}{2} \\
&\le D(2D^2+1)\sum_i \mathbb{E}|X_i|^4.
\end{aligned}$$